



Machine Learning II

Data Science M.Sc.

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Classification

Maximal Margin Classifier & Support Vector Classifier



Introduction

A support vector machine (SVM) is a **Supervised Learning** method.

- ▶ It is a **classification** method with K classes.
- ▶ Usually an SVM is applied as **binary classifier** so $K=2$.
- ▶ Extending to $K > 2$ is not trivial, we will consider this later.



The method's name

Q: Where does the strange name come from?

- ▶ **Vector:** Here a vector is a data point, an element (row) in a data set with numeric values on p -variables.
Element of a p -dim vector or feature space, where each observation is a data point.
- ▶ **Support:** Some data points (vectors) define (support) the boundary between two classification regions.
- ▶ **Machine:** As in machine learning: a prediction model.



Comparison to LDA

- ▶ In linear discriminant analysis (LDA) we partitioned the feature space into two classes separated by a straight line.
- ▶ The SVM classifies a similar type of classification by identifying a boundary between feature space regions, but uses a very different approach.



Maximal Margin Classifier

Let's start simple:

- ▶ For ease, we will assume at first that the data have two predictor variables and one outcome variable.
- ▶ Therefore, *vectors* are points in $(x_1, x_2) \in \mathbb{R}^2$, and each point belongs to one of $K=2$ classes labelled either -1 or $+1$.
- ▶ We choose an arbitrary line and assign one class to each side of that boundary.
⇒ It is very likely that there are many misclassifications.
- ▶ We can rotate and shift the line so that the number of misclassifications decrease.
If we are very lucky there is a boundary which gives no misclassifications.
⇒ This means the two classes would be totally separated.



Defining the linear boundary

Suppose we have a straight line with the form $x_2 = a + b \cdot x_1$, we can express this line the form¹

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 = 0$$

Now consider 3 points $\mathbf{x}_i = (x_{i1}, x_{i2})$:

$$(x_{11}, x_{12}) \text{ lies on the boundary} \Leftrightarrow x_{12} = a + b \cdot x_{11} \Leftrightarrow \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} = 0$$

$$(x_{21}, x_{22}) \text{ lies on one side of the boundary} \Leftrightarrow x_{22} > a + b \cdot x_{21} \Leftrightarrow \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} > 0$$

$$(x_{31}, x_{32}) \text{ lies on the other side of the boundary} \Leftrightarrow x_{32} < a + b \cdot x_{31} \Leftrightarrow \beta_0 + \beta_1 x_{31} + \beta_2 x_{32} < 0$$

This gives an easy classification rule:

$$y = f(x_{i1}, x_{i2}) = \begin{cases} +1 & \text{if } \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} > 0 \\ -1 & \text{if } \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} < 0 \end{cases}$$

¹Note that this expression is over-parametrised. If you multiply all three β s by a constant, the same boundary is obtained. Later we will use a normalized version so that $\beta_1^2 + \beta_2^2 = 1$.



High dimensional feature spaces

The definition of the boundary line on the previous slide generalises into higher dimensional feature spaces, i.e. $\mathbf{x}_i = (x_{i1}, \dots, x_{ip}) \in \mathbb{R}^p$.

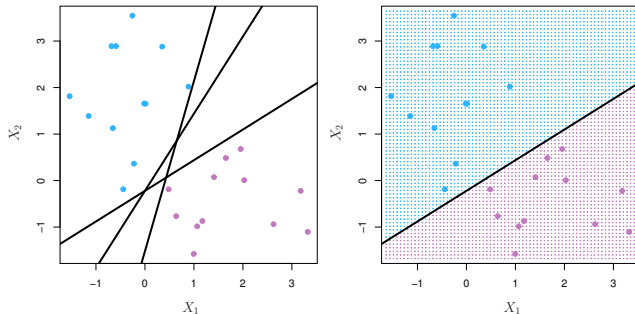
$$y = f(\mathbf{x}_i) = \begin{cases} +1 & \text{if } \beta_0 + \sum_{j=1}^p \beta_j x_{ij} > 0 \\ -1 & \text{if } \beta_0 + \sum_{j=1}^p \beta_j x_{ij} < 0 \end{cases}$$

- ▶ If each data point is a 3-dimensional vector $\mathbf{x}_i = (x_{i1}, x_{i2}, x_{i3})$ then the boundary is a **plane** defined by $\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 = 0$.
- ▶ In higher dimensions the boundary is called a **hyperplane**.
- ▶ From now on we will use the term hyperplane for the boundary, even if the data are 2 or 3 dimensional.
- ▶ In this lecture, we only consider $\mathbf{x}_i \in \mathbb{R}^2$. The adaptation into higher dimensions is straight forward.



Ambiguity of hyperplane solutions

If our data are separable, then there will infinitely many hyperplanes which are perfect classifiers (left diagram).



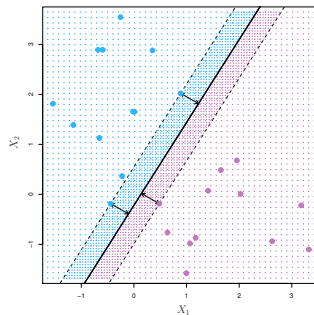
Q: Do we care which hyperplane to use?

A: Yes, because we want to use the method to predict the class for test/future data.



Optimal hyperplane solution

The **maximal marginal hyperplane** is the hyperplane which is the furthest away from its nearest data points. There is a strip either side of the boundary which contains no points, this is called the *margin*.





Maximal marginal hyperplane

Let M be the distance between the boundary and margin edge, i.e. half the margin width. We want to maximise M .

- We require that every point is on the correct side of the hyperplane, so:

$$\text{For } y_i = +1 \quad \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} > 0$$

$$\text{For } y_i = -1 \quad \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} < 0$$

- These two rules can be expressed as one rule:

$$y_i \cdot (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) > 0$$

Theorem:

Under the assumption² that the normal vector of the 2-dim hyperplane (β_1, β_2) is a unit vector having length 1, i.e. $\beta_1^2 + \beta_2^2 = 1$, then the distance from the line $\beta_0 + \beta_1 x_1 + \beta_2 x_2 = 0$ to the point (x_{i1}, x_{i2}) is equal to

$$y_i \cdot (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}).$$

Proof: https://en.wikipedia.org/wiki/Distance_from_a_point_to_a_plane#Closest_point_and_distance_for_a_hyperplane_and_arbitrary_point

²see footnote on slide 7.



Margin classifier optimization

To ensure that every point is at least M units away from the boundary, we require

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) \geq M \quad \text{and} \quad \beta_1^2 + \beta_2^2 = 1.$$

Optimization criteria:

Find the *maximin*³ solution for $y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2})$:

$$M = \max_{\beta_0, \beta_1, \beta_2} \left\{ \min_i \{y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2})\} \right\}, \quad \text{subject to } \sum_{j=1}^2 \beta_j^2 = 1.$$

³<https://en.wikipedia.org/wiki/Minimax>



Margin classifier algorithm

In terms of developing the algorithm it is better to re-express the algorithm as

- ▶ Choose $\beta_0, \beta_1, \beta_2$ to maximise M , such that
- ▶ $y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) \geq M$ for $i=1, \dots, n$,
- ▶ subject to the constraint $\sum_{j=1}^2 \beta_j^2 = 1$.

The size of the maximal margin can be used as a measure of how confident we are in the maximal margin classifier.

Remark:

- ▶ This is the format in James et al.
- ▶ Many other authors use an alternative equivalent or dual version:

$$\text{minimise } \sum_{j=1}^2 \beta_j^2 \text{ subject to } y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) \geq 1.$$



Comments

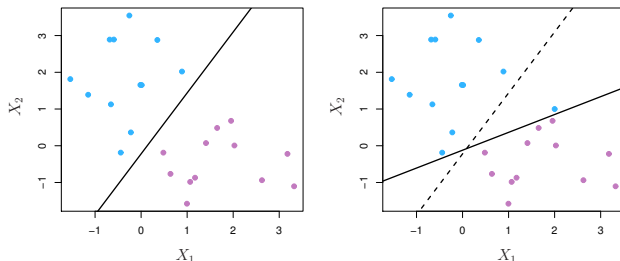
- ▶ There are **at least** three data points that lie on the edge of the margin, their distance from the optimal hyperplane is M . These points are called the **support vectors**, they *support* the hyperplane.
- ▶ If you remove a non-support vector from the dataset, the optimal hyperplane won't change.
- ▶ Once the best hyperplane has been found, the **predictions are easy to obtain** by taking the sign of the hyperplane function.
I.e. for a test observation vector $\mathbf{x}_t$

$$\hat{y}_t = \text{sign}(\beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2}) = \begin{cases} 1 & \text{when } \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} \geq 0 \\ -1 & \text{when } \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} < 0 \end{cases}$$



Support vector classifiers and reality I

Even if there is a hyperplane that separates the data perfectly, the resulting regions can be quite sensitive to values near the hyperplane.

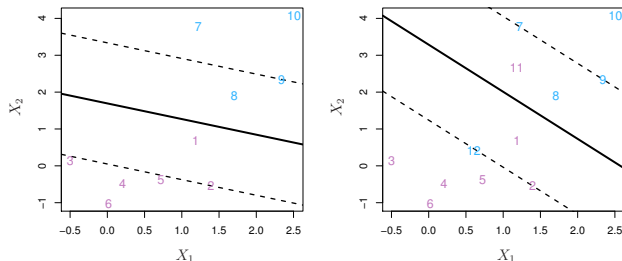


The image on the right is the same as on the left except for the addition of one(!) new blue point.



Support vector classifiers and reality II

Usually we don't have two perfectly separated classes.



The *support vector classifier* allows for some points to be in the margin or on the wrong side of the border, the hyperplane has a *soft margin*.



Support Vector Classifier with soft margins

The adaptation from the previously introduced algorithm is straight forward:

We introduce *slack variables* $\epsilon_i \geq 0$ for each observation $\epsilon_1, \dots, \epsilon_n$, which allow some of the points to move into the margin.

- ▶ $\max_{\beta_0, \beta_1, \beta_2, \epsilon_1, \dots, \epsilon_n} M$, such that
- ▶ $y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) \geq M(1 - \epsilon_i)$ for $i=1, \dots, n$,
- ▶ subject to the constraints $\beta_1^2 + \beta_2^2 = 1$, $\epsilon_i \geq 0$ and $\sum_{i=1}^n \epsilon_i \leq C$.

C is a positive *hyperparameter*.

We consider, that it could be worthwhile to misclassify a few training observations in order to do a better job in classifying the remaining observations.



Properties of slack variables

If ϵ_i is ...

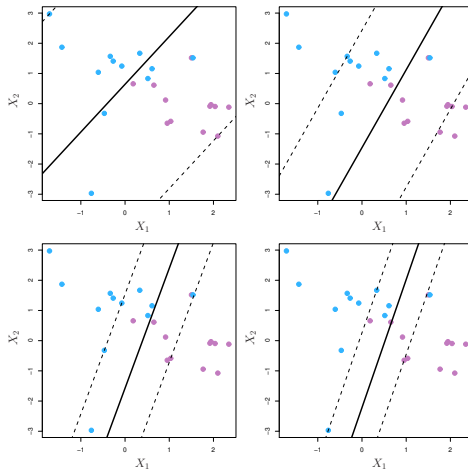
- ▶ $\epsilon_i = 0$, the i -th data point is on the correct side of the margin.
- ▶ $0 < \epsilon_i < 1$, the i -th data point is in the margin, but on the correct side of the hyperplane.
- ▶ $1 < \epsilon_i < 2$, the i -th data point is in the margin, on the wrong side of the hyperplane.
- ▶ $\epsilon_i > 2$, the i -th data point is outside the margin, on the wrong side of the hyperplane.

C can be considered as a budget, every point that crosses the margin has a non-zero cost ϵ_i and the total cost are not allowed to exceed C .

- ▶ If $C=0$, we return to the maximal margin classifier.
- ▶ The being on the wrong side of the hyperplane costs at least 1 unit. If we want no more than 5 misclassifications in the training data, we can start by setting $C=5$.



Budget hyperparameter C



By decreasing C the width of the margin decreases.



Final remarks

- ▶ Predicted values from a support vector classifier are obtained in exactly the same way as for the maximal margin classifier.
- ▶ Now, all the points which lie in the margin or on the wrong side are the support vectors. They all contribute to defining the parameters for the optimal hyperplane.
- ▶ As before removing a non-support vector has no influence on the hyperplane.
- ▶ Moving a non-support vector has no influence on the hyperplane unless it crosses into the margin.
- ▶ This property means that extreme values on the correct side of the hyperplane do not affect the classifier, meaning that the classifier is moderately robust.
- ▶ LDA gives similar results but is influenced by extreme values in all directions. Why?